

HROC • NCAN • Deep-Learning Track

Does the cortex change as the spinal cord learns?

First real-data results — Animal 9 through the full pipeline, end to end.

Suchith Rao • prepared for Dr. Carp • July 2026



The question, in one line

The paradigm

1

Baseline

Record the spinal H-reflex (a leg-muscle reflex) without training.

2

Down-conditioning

Reward the animal for making the reflex smaller. The spinal cord slowly learns.

3

The brain?

We record cortex (ECoG) at the same time — and ask if IT changes too.

What's new here

Chad already published the simple link — when the reflex goes down, ECoG activity shifts. That correlation is done.

Our new question:

Does the brain's internal representation drift across conditioning — a cortical fingerprint of learning that's never been shown?

THIS WEEK

From a synthetic demo to real data

BEFORE

The pipeline ran on synthetic (fake) data shaped like Animal 9 — enough to prove the machine works, but not a scientific result.

NOW

The exact same pipeline is trained on the real Animal 9 recordings — decoded, phase-labeled, and run end to end.

283,315

real trials decoded

3

conditioning phases recovered

2

brain + muscle channels

ORIENTATION

What one trial actually is

Animal 9's recordings live in a 2006 lab database. Each stimulus creates one trial — a 30-millisecond snapshot, sampled 5,000 times a second, on two electrodes at once:



Muscle — EMG (soleus)

The leg-muscle response to the stimulus. This is where the **H-reflex** lives — the thing conditioning trains.

→ *our prediction target*



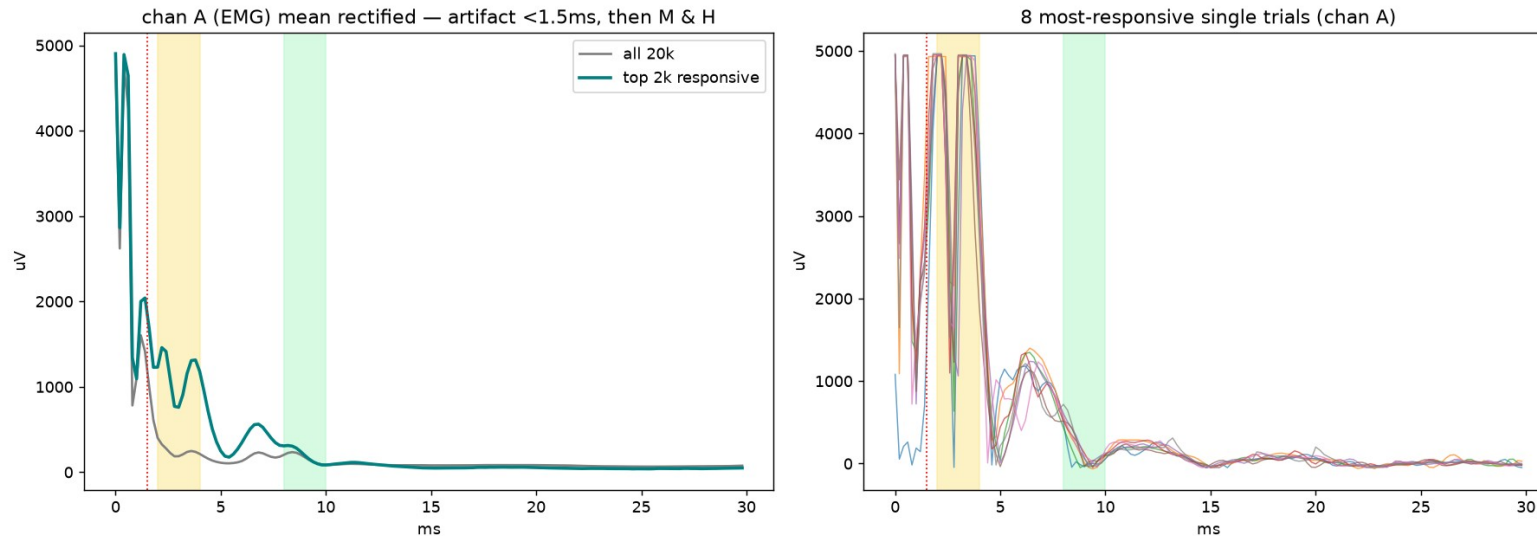
Brain — ECoG (cortex surface)

Electrical activity from the surface of the motor cortex, recorded at the very same moment as the muscle. This is the brain signal we study.

→ *the model's input*

STEP 1 OF 3 • GETTING TO REAL DATA

First: prove we're reading the signal right



What this took

- Loaded the 2.3 GB database locally
- Cracked the byte format (it was mislabeled in the lab docs)
- Confirmed channel A = muscle, B = brain
- Verified against textbook M / H timing

The raw data is a 20-year-old binary format. We confirmed the decode by averaging the muscle channel: it shows the textbook shape — a stimulus artifact, then the M-wave (2–4 ms), then the H-reflex (~6–7 ms). Real physiology, not noise.

Recover the conditioning phases

We read the experimenter's own typed log from 2006 to find when each phase began — then tagged all 283k trials by time.

Baseline

149,064

Before conditioning onset

Down-cond early

71,743

Conditioning underway

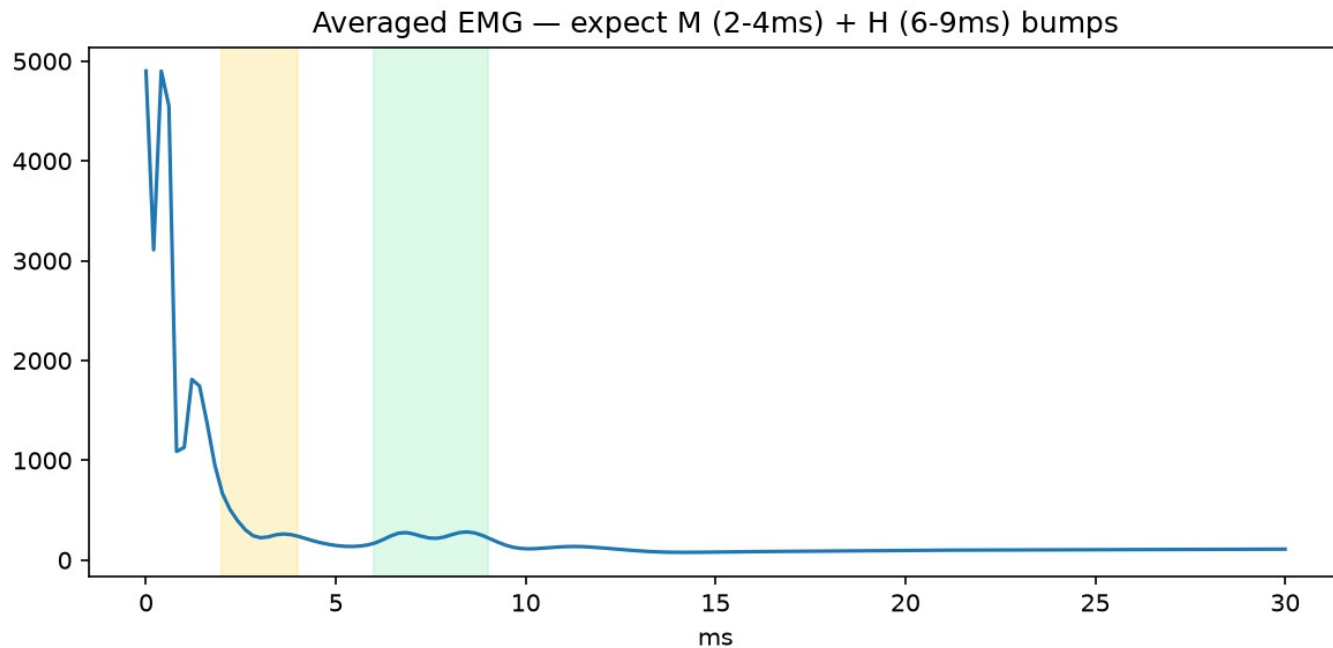
Down-cond late

62,508

Effect fully developed

Corrected boundaries: the log's “stopped conditioning” note is misleading — the reflex keeps dropping long after it, so conditioning actually continued. We anchor the boundary on the documented conditioning ONSET and split early/late by time.

Define what we predict: the H-reflex size



The measure

Where: 6–9 ms — the experimenter's log says exactly that, and it matches what we measured.

How: mean rectified amplitude in that window — your spec, not a single fragile peak.

Green band = the H-reflex window on the real grand-average.

A 'cortical fingerprint' of each trial

Instead of hand-picking features, the model learns its own compact summary of the brain signal — call it a fingerprint (48 numbers per trial). It's trained in two stages:

1 Learn the fingerprint

The model compresses each brain trace down to 48 numbers, then rebuilds the original from them. To succeed, those numbers must capture the real structure of the signal.

No labels used here — it never sees the H-reflex.

2 Test what it captured

We freeze the fingerprint so it can't change, then attach a small predictor and ask: can you read the leg's H-reflex size from the brain fingerprint alone?

If yes → the brain signal genuinely carries H-reflex information.

How to read the next three slides

● The fingerprint

48 numbers the model invents to summarize each brain trace. Nobody tells it what to look for — it finds its own summary.

● R^2 (a 0-to-1 score)

How much of the reflex the fingerprint can explain. 0 = nothing, 1 = perfect. After controlling for the stimulus it's ~0.04 — small but real.

● Drift

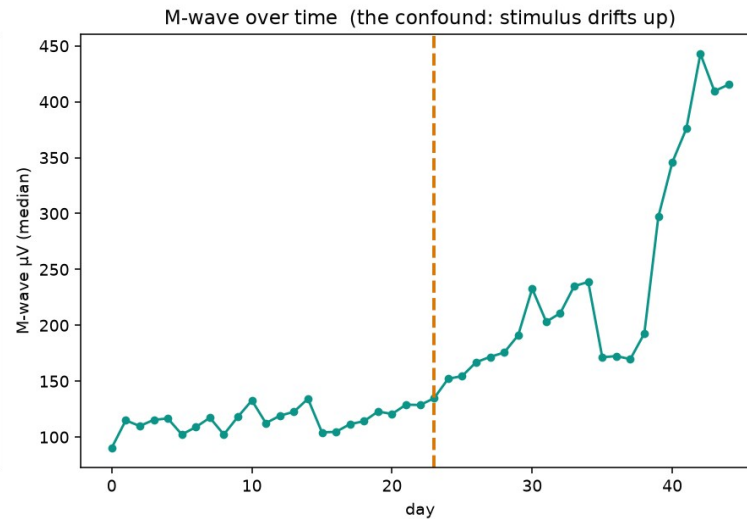
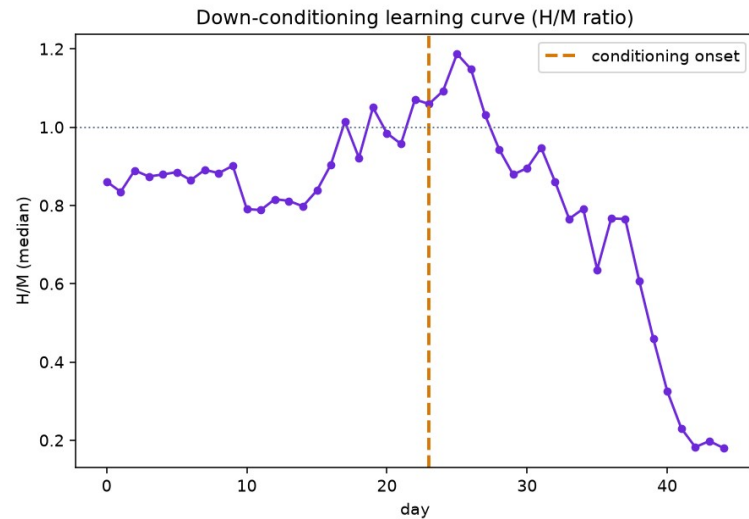
How far the average brain state has moved from where it started at Baseline. A bigger number means the representation changed more.

● The M-wave control

The reflex depends on stimulus strength. We divide by (or remove) the M-wave everywhere, so we measure learning, not stimulus.

RESULT 1 • THE BEHAVIOUR IS REAL

Down-conditioning worked — the textbook curve



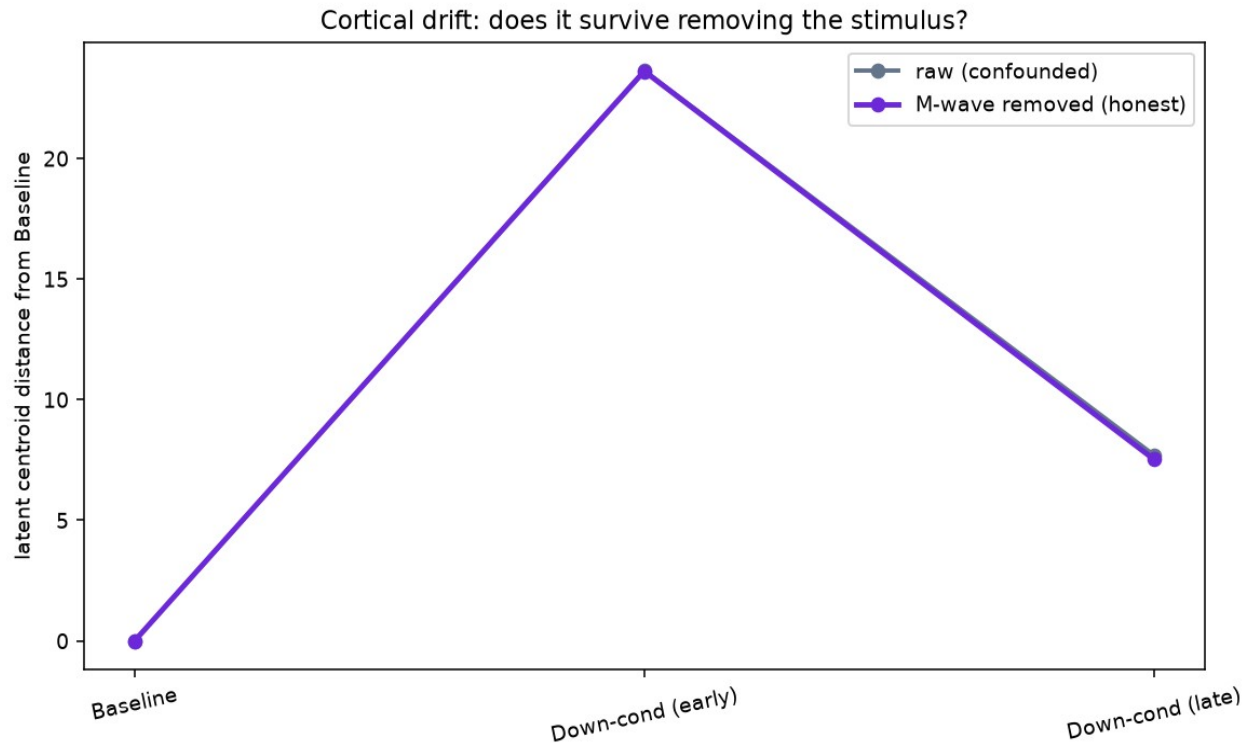
Why H/M, not raw H

The reflex (H) depends on how hard you stimulate. So we divide by the M-wave — the direct-muscle response in the same trace — exactly as the Wolpaw lab does.

H/M falls 1.17 $\rightarrow$ 0.59 (median down to 0.18). Right panel: the stimulus drifted UP over the same weeks — which is exactly why raw H is invalid.

RESULT 2 • THE TEST THAT SETTLES IT

Does the cortical drift survive the stimulus?



The critical control

We mathematically remove the M-wave (stimulus) from the brain fingerprint, then re-measure the drift.

The two lines barely move apart. The cortical drift is NOT a stimulus artifact — it's real.

Caveat: the shift is biggest at conditioning onset, not steadily growing — cortex reacts to the new task, doesn't simply mirror reflex size.

Can the brain predict the reflex? A little — honestly

We test how much of the reflex each source explains, using the same M-wave control. Three numbers, side by side:

0.20

**Stimulus alone
(M-wave → reflex)**

The stimulus explains most of it. This is the confound.

0.05

**Cortex, raw
(fingerprint → reflex)**

A weak link on its own.

0.04

Cortex, stimulus removed

Survives the control — small but genuinely there.

The earlier $R^2 \approx 0.17$ was inflated — it rode on shared stimulus variance. The honest cortex-only contribution is ~ 0.04 .

Is this good progress? My honest scorecard

STRONG

Infrastructure + behaviour

Real data end to end, and the behaviour is textbook: down-conditioning worked (H/M fell to ~ 0.18). Reproducible on a laptop.

REAL BUT MODEST

Cortical effect

The cortical drift survives removing the stimulus — not an artifact. But the cortex $\rightarrow$ reflex link is small (~ 0.04) and onset-driven.

NOT YET

A publishable claim

Needs more animals, up-conditioning controls, and statistics on the drift. One animal, one direction — a strong foundation, not a result.

NEXT

The plan — and it scales

Animal 9 is 1 of ~14 conditioned animals sitting in the Drive. The pipeline is built to pool them — every new animal makes the drift result stronger.

1

Resolve the baseline confound

Restrict 'Baseline' to post-setup trials; regress out raw reflex size. Does the drift survive?

2

Add statistics

Bootstrap the drift — a within-baseline split should drift ~0. Is 28.5 vs 5.0 real?

3

Pool more animals

Convert 6, 10, 11, 12, 13... one at a time, watching for animal-identity confounds.

4

Bring in controls

Up-conditioning animals should drift the other way; freely-running should barely drift.

WHERE I NEED YOUR READ

Four neuro calls — not code

1 Baseline definition

Given the setup-period confound, how should we define a clean behavioral baseline for Animal 9?

2 The H-reflex label

Confirm: mean rectified amplitude, 6–9 ms window — is that the measure you want us predicting?

3 What would convince you

What controls make the cortical drift believable — up-conditioning reversing it? freely-running as a null?

4 Which animals next

Of the ~14 in the Drive, which should we prioritize converting and pooling?